DP Biology – D4.3 (HL extension)

Timing, life cycles and evolution

- What is driving climate change?
- How does climate change affect ecosystems?

HL statements D4.3.9–12. Textbook pp. 834–837.
Image: a tawny owl. Akumiszcza, CC BY 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	D4.3.9–10	Explain what controls the timing of biological events, and how climate change disrupts it
2	D4.3.11	Explain why warming gives insects more life cycles in a year
3	D4.3.12	Explain evolution in response to climate change

The big picture: timing matters

Phenology	Mismatch	Response
When events happen	Partners drift out of step	Faster insects; evolving species

Species time events with **day length** or **temperature**. Warming shifts temperature cues but not day length, so partners drift apart.

Big question

Why are great tit chicks hatching too late for their caterpillar food?

Phenology

Phenology is the study of the timing of biological events: flowering, budburst, bud set, migration, nesting.

Long records show many spring events arriving **earlier**.



Flowering time is a phenological event. Phil aplosh, CC BY 2.0.

Do it

Record the budburst date of one school tree each year.

Two main cues

D4.3.9

Photoperiod (day length)

Set by date and latitude: does NOT change with climate

Temperature

Changes with warming: events come earlier

Day-length examples	Temperature examples
Bud set; timing of migration	Budburst; insect development

Key idea

If partners use different cues, warming moves one but not the other.

How scientists track timing

D4.3.9

Source	Example
Historical records	Kyoto cherry blossom dates for over 1000 years
Citizen science	First swallows, frogspawn, leaf burst
Satellites	When large areas turn green
Nest boxes	Great tit laying dates and caterpillar peaks

Nature of science

Long, consistent records are needed to separate a trend from year-to-year noise.

Out of step: reindeer and Arctic plants

D4.3.10



Arctic plants, Alaska. BLM Alaska, public domain.

Plants grow **earlier** as it warms (temperature cue).

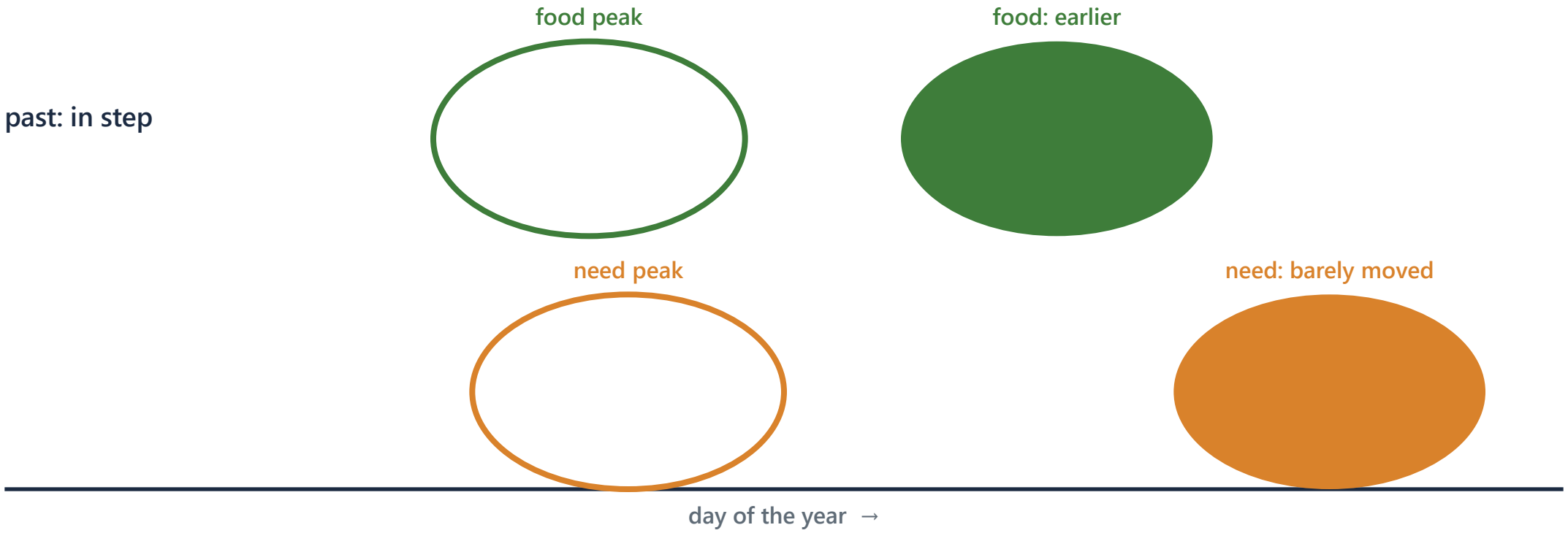
Reindeer migrate by **day length**, arriving on the same date, after the most nutritious growth: fewer calves survive.

Think

Why does warming matter more for an animal using day length than for one using temperature?

Drifting apart

D4.3.10

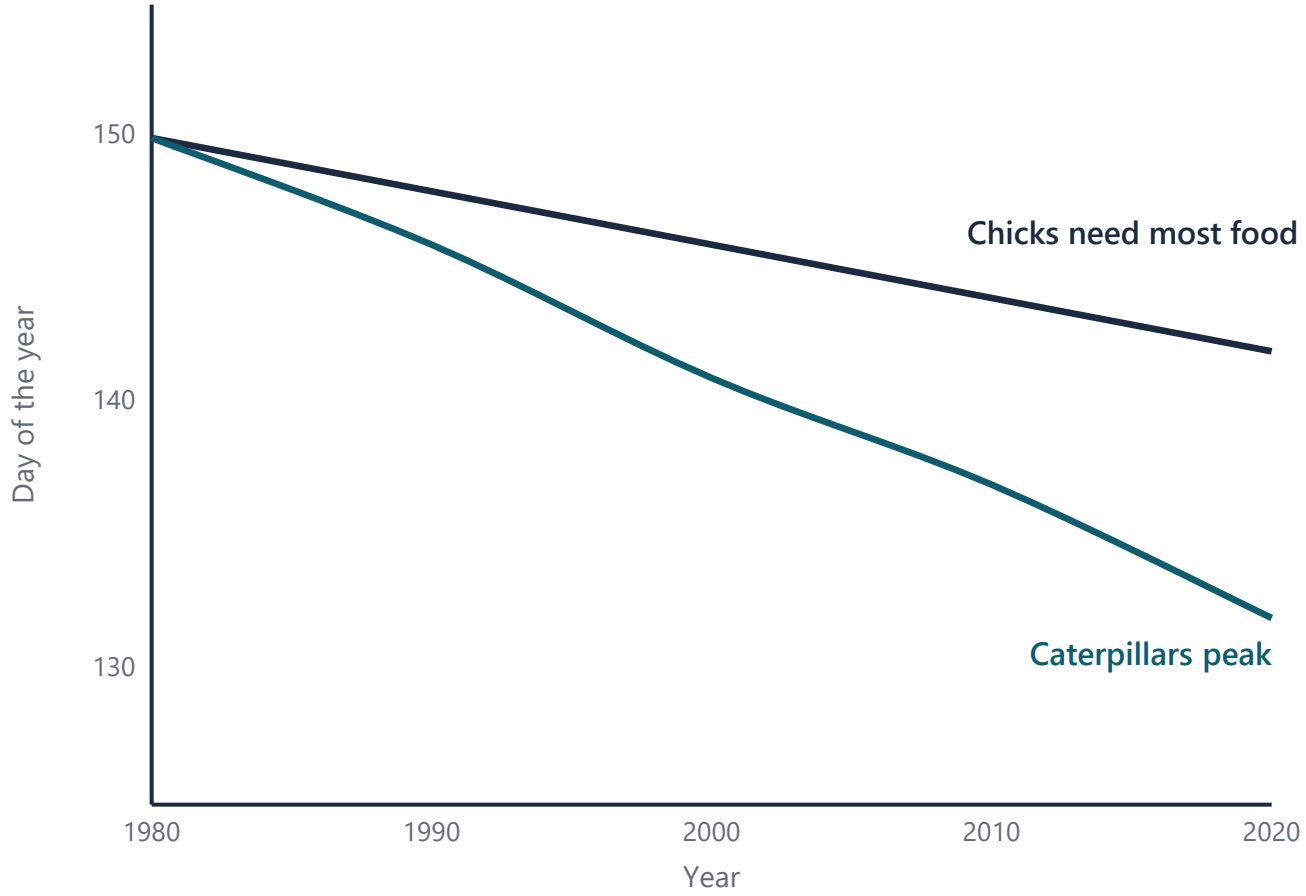


Key idea

When food and need peaks separate, young are poorly fed and survival falls.

Out of step: great tits and caterpillars

D4.3.10



Schematic trends, not measured values.



Great tit. Dan Davison, CC BY 2.0.

Caterpillars track warmth; great tits shift laying less, so chicks hatch after the food peak.

Check-in: class 1

D4.3.9–10

1. Define phenology.
2. Name the two main timing cues.
3. Explain why day-length cues cause mismatch.
4. Outline the reindeer example.
5. Outline the great tit example.

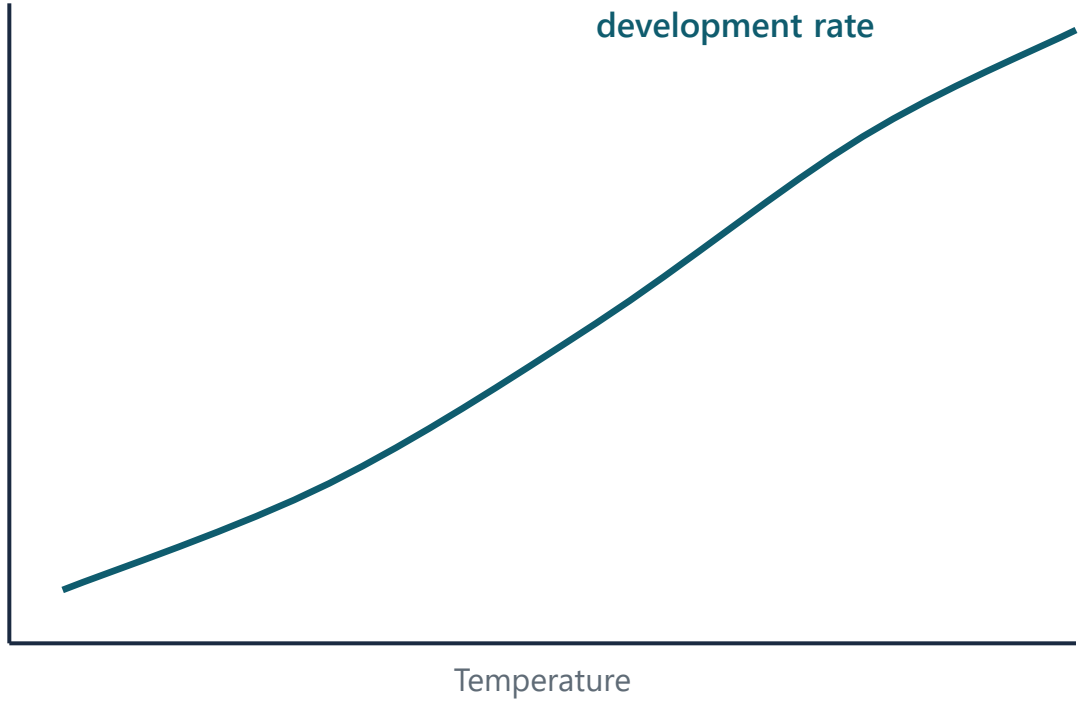
How to use this

Answer without your notes, then check against slides 3–9.

Warming speeds insect life cycles

D4.3.11

Insects are ectotherms: they develop **faster** when warm. A longer, warmer season fits **more generations** into one year.

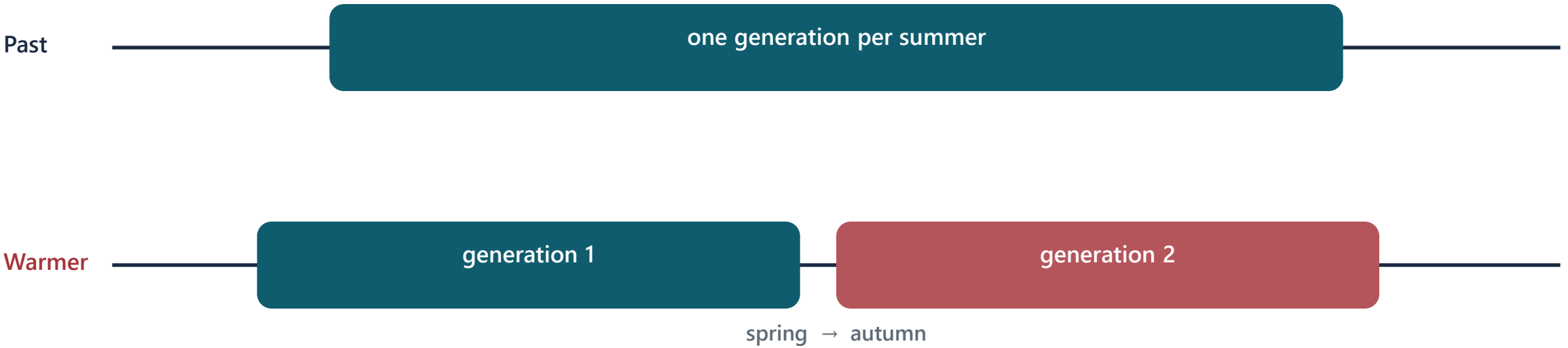


So

More generations = more insects by late summer, with more chances to spread.

One generation or two?

D4.3.11



Key idea

Two generations can double the population reached by autumn.

Spruce bark beetles

D4.3.11



Spruce bark beetle, Ips typographus. Daniel Adam, Bugwood.org, CC BY 3.0 us.

Northern populations once had **one** generation a year; warming allows **two**, and they spread further north.

Warmer summers → Two generations → More beetles per tree → Defences overcome → More spruce die

Working with data: beetle damage

D4.3.11

Spruce killed by bark beetles in a region (thousand m^3) and mean summer temperature. Invented values; realistic pattern.

Year	Summer mean / °C	Spruce killed
2000	15.1	40
2010	15.8	90
2018	17.2	600
2020	16.5	380

1. Describe the relationship.
2. Explain the 2018 peak.
3. Suggest one other factor.

Check-in: class 2

D4.3.11

1. Explain why warmth speeds insect development.
2. Explain how warming gives more generations.
3. Outline the spruce bark beetle example.
4. Explain why trees are overwhelmed.
5. Link beetle damage to climate feedback.

How to use this

Answer without your notes, then check against slides 11–14.

Evolution: tawny owl colour

D4.3.12



Tawny owl. Akumiszcza, CC BY 3.0.

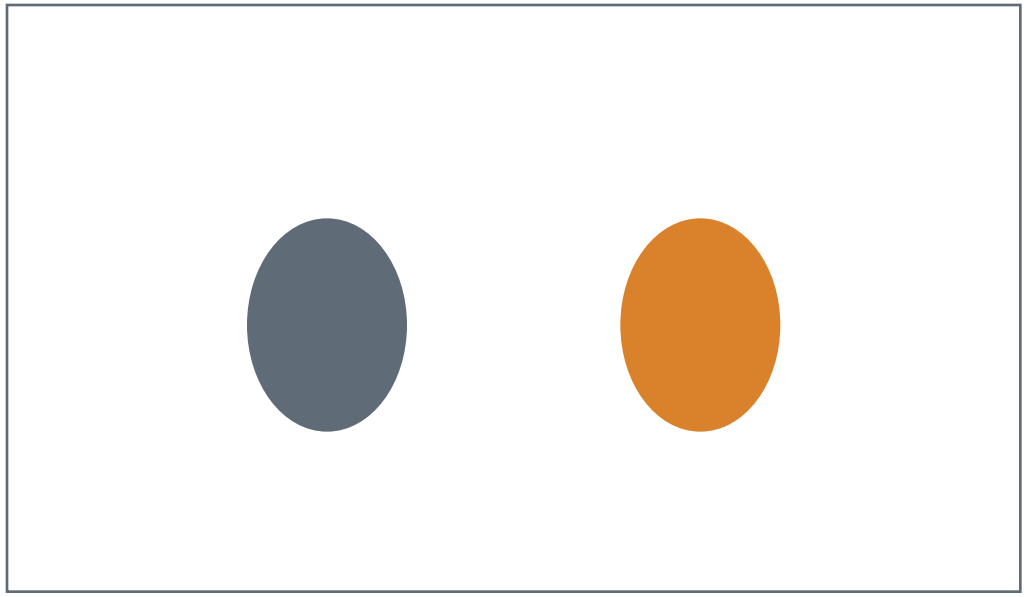
Tawny owls are pale grey or brown; colour is **heritable**.

Grey owls hid best in snowy Finnish winters. As snow cover fell, brown owls rose from about 30% to about 50%.

Less winter snow → Grey owls more visible → Grey owls survive less → Brown allele rises

Snow and owls

D4.3.12



snowy: grey hidden, brown seen



snow-free: brown hidden, grey seen

Key idea

A changed climate changes which variant has the highest adaptive value.

Evolution: bud set in Scots pine

D4.3.12

Bud set is when a tree stops growing and forms winter buds. Its date is partly heritable.

Too early wastes growing time; too late risks frost. A longer season favours **later** bud set.

Prediction

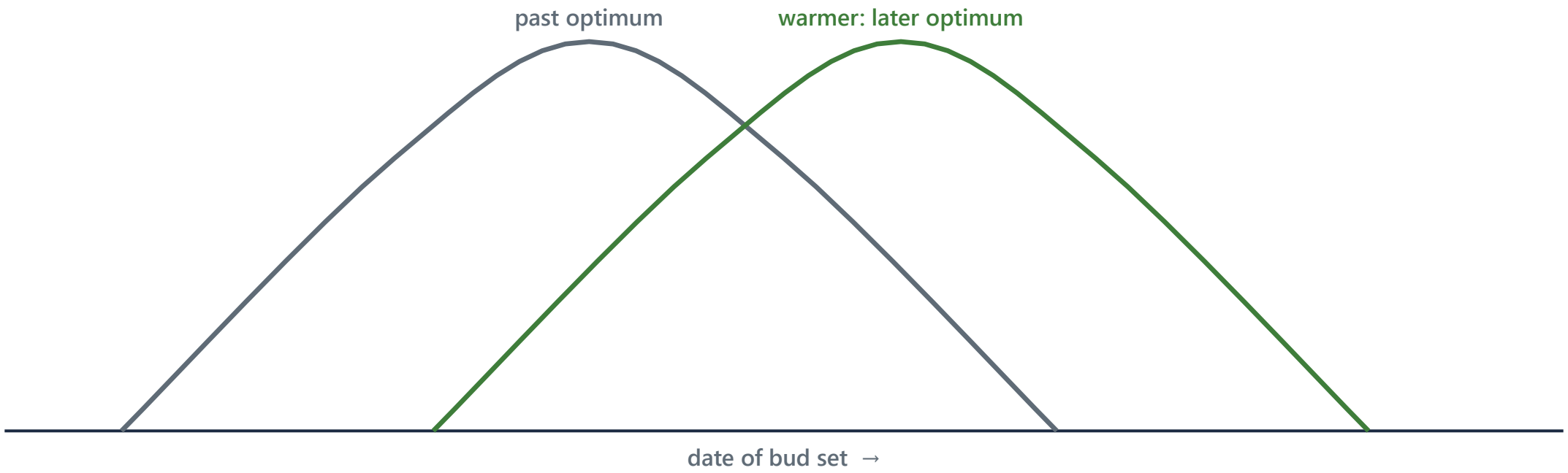
Over generations, mean bud set should move later as alleles for later bud set rise.

Limits

Trees have long generations, so evolution may be too slow to keep up.

A shifting optimum

D4.3.12



Read it

Curves show fitness; the best date shifts later, so selection favours late setters.

Practice: timing and evolution

D4.3.9–12

Explain

Why the brown owl allele is rising.

Predict

Bud set date in 100 years.

Evaluate

Can trees evolve fast enough?

Use

heritable, adaptive value, allele frequency, generation time

Working with data: caterpillars and chicks

D4.3.10

Day of year of the caterpillar peak and great tit hatching peak. Invented values; pattern matches long-term studies.

Decade	Caterpillar peak	Hatching peak
1970s	150	148
1990s	142	145
2010s	135	141

1. Calculate the mismatch each decade.
2. Explain the change.
3. Predict the effect on great tits.

Check-in: class 3

D4.3.12

1. Explain the tawny owl example.
2. State what makes evolution possible here.
3. Define bud set.
4. Predict how bud set will evolve.
5. Explain why some species may not evolve fast enough.

How to use this

Answer without your notes, then check against slides 16–21.

Common mistakes to avoid

Not this	But this
All species shift timing equally.	They respond differently: mismatch.
Day length changes with climate.	It depends only on date and latitude.
Owls choose to change colour.	Morph frequencies change by selection.

Summary: timing, life cycles and evolution

D4.3.9–12

	Key idea
D4.3.9	Phenology; photoperiod and temperature cues
D4.3.10	Mismatch: reindeer, great tits
D4.3.11	More insect generations: bark beetles
D4.3.12	Evolution: tawny owls, Scots pine bud set

Check that you can...

1. **Define** phenology and **outline** timing cues
2. **Explain** mismatches with examples
3. **Explain** faster insect life cycles
4. **Outline** evolution in response to climate change

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Define phenology.
2. Explain why climate change can put two species out of step.
3. Explain why the proportion of brown tawny owls has increased.

Image credits

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Boreal Forest Flora (53451032212).jpg – BLM Alaska – Public domain

Great tit.jpg – Dan Davison – CC BY 2.0

Ips typographus.jpg – Daniel Adam, Office National des Forêts, Bugwood.org – CC BY 3.0 us

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